- Introduction
- Full-Custom (VLSI) IC Technology
- Semi-Custom (ASIC) IC Technology
- Programmable Logic Device (PLD) IC Technology

8.1 Introduction

A structural representation of the system generally deals with the various components and their interconnections to implement system's functionality. IC technology is more about mapping the structural representation to a physical implementation. The physical implementation can be done using various methods, out of which full-custom, semi-custom and programmable technologies are few common methods. As CMOS transistor is the core of every component, let us take a look at CMOS transistor and different layers required for its physical implementations.

CMOS Transistor

CMOS transistor consists of three terminals: the source, drain, and gate. Source and drain are created by implanting ions on the surface of silicon. Gate is formed using poly-silicon, and lies between source and drain. Gate is placed on top of silicon and is isolated from silicon with the help of silicon dioxide insulating layer. Gate voltage controls the current flowing from source to the drain. In case of nMOS transistor, a high voltage at gate will attract electrons from silicon substrate towards it resulting in formation of conducting channel between source and drain. For a low voltage at gate, the conducting channel is not formed.

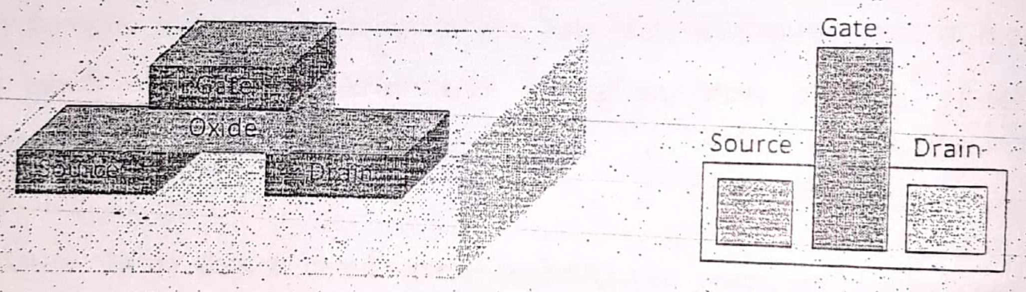

Figure 8.1: CMOS Transistor and Its Top-Down View

Layers in Physical Implementation

The transistor basically has three layers: diffusion layer for source and drain, oxide layer for insulation, and poly-silicon layer for gate. For circuits, there will be number of transistors connected together to represent particular functionality. These connections are represented by metal layers. There can be number of metal layers based on complexity of circuit implemented. Each metal layer is insulated from another layer using oxide layer. Hence, there exists number of oxide layer.

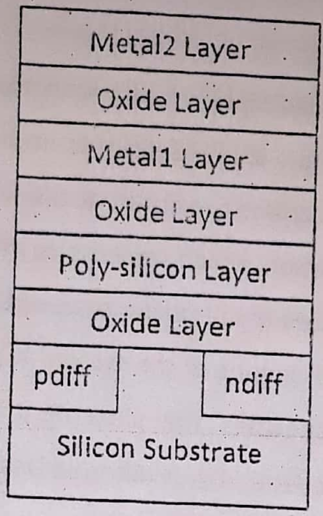

Figure 8.2: Basic Layers in Physical Implementation

Example 1: Draw the transistor level circuit schematic and top-down view for a NAND gate

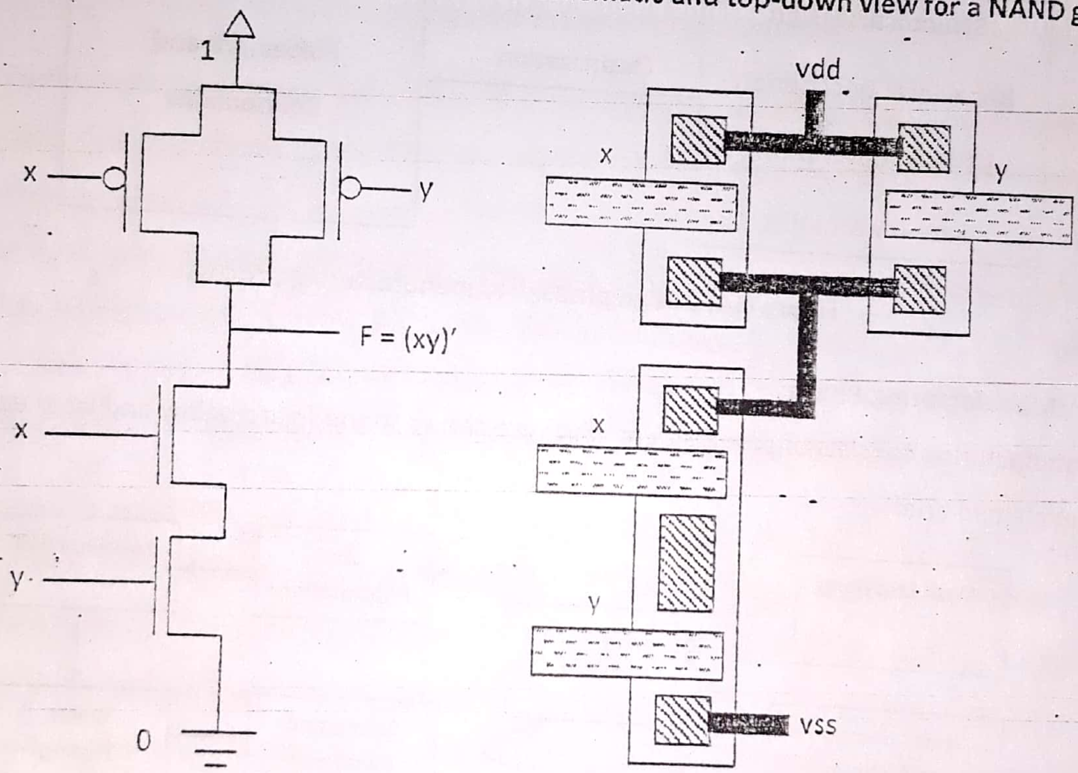

Figure 8.3: Circuit Schematic and top-down view of NAND gate

IC Manufacturing Process

Basically, IC manufacturing process can be divided into two phases: design phase and manufacturing phase. In design phase, structural design and layout design is done, whereas manufacturing phase includes various steps from mask creation to final IC packaging.

A. Design Phase

In design phase, the structural description along with the layout of the system is developed. Initially, the behavioral description of the system is implemented using hardware description language. The high-level HDL describes the circuit at the Register Transfer Level. The first step in the synthesis process is compilation which converts high-level VHDL language into a netlist at the gate level. The second process is speed and area optimization which is performed on gate-level netlist. Finally, the physical layout of the system is generated with the help of place-and-route software. The layout specifies the placement of every transistor and every wire connecting those transistors. Several EDA (Electronic Design Automation) tools are available for circuit synthesis, implementation, and simulation.

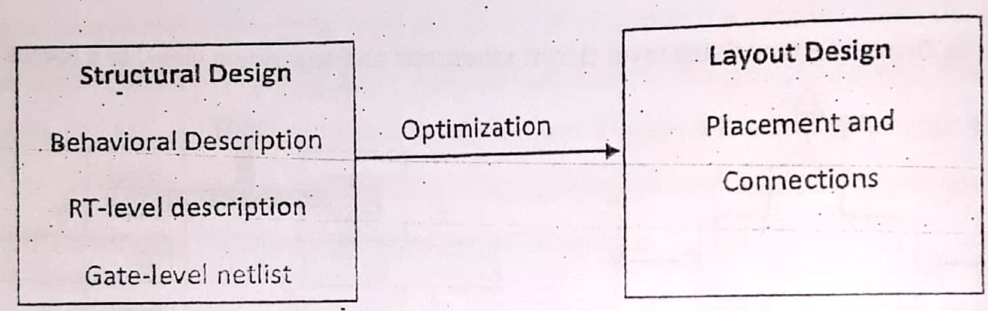

Figure 8.4: Design phase in IC manufacturing process

B. Manufacturing Phase

Manufacturing consists of several steps which are shown in the figure below and later each step is explained briefly.

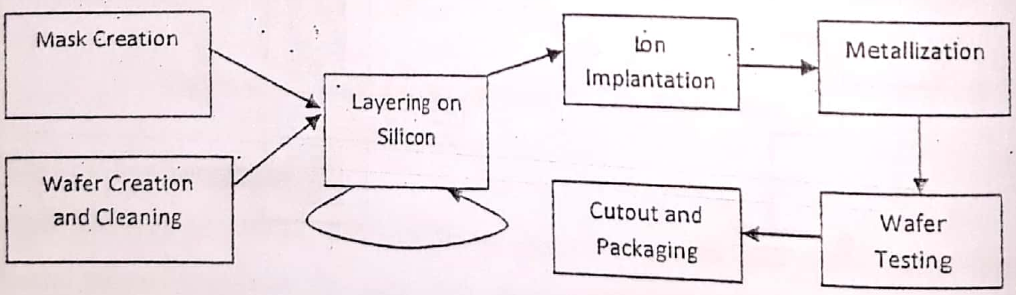

Figure 8.5: Manufacturing phase in IC manufacturing process

Mask Creation: The layout design of the system is translated into masks. The number of masks requirement may vary based on number of layers defined by the systems complexity. Masks for different layers – such as oxide layer, metal layers, etc – are generated. Generally, masks contain number of identical regions, so that number of IC's can be produced at once.

Silicon Wafer Creation and its cleaning: In a crucible, high purity silicon is melted. Donor impurity atoms can be added to dope the crystal. A seed crystal is dipped into molten silicon and pulled upwards rotating it. And cylindrical ingot is extracted by controlling temperature gradients, rate of pulling and speed of rotation. Finally, the ingot is sliced with a wafer saw and polished to form wafers.

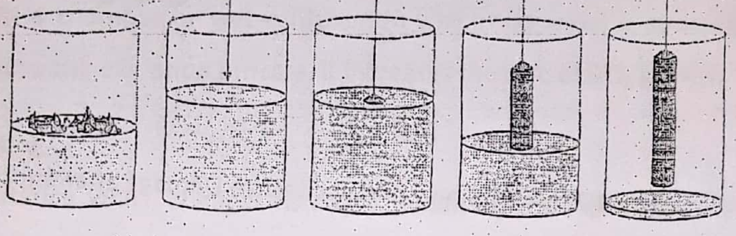

Figure 8.6: Silicon Wafer Creation

Wafer must be cleaned before any layer is deposited on it. Various cleaning methods can be used. Chemical cleaning methods are commonly used. First method of chemical cleaning is by using piranha solution in which wafer is immersed in hot mixture of hydrogen peroxide and sulfuric acid. Another method is using sonic waves in cleaning solution which is known as megasonic cleaning process. After the water is cleaned with chemical, it must be rinsed with De-Ionized (DI) water. Finally, the wafer is dried using either nitrogen gun or by baking. Also spun dry method can be used to make the wafer dry after cleansing process.

Layering on Silicon: Various layers are developed on the silicon surface. Layer for masks can be created using different layering techniques. Photolithography, which uses optical radiation to create patterns, is very common method in layering process. In this process, the layer required, for example silicon dioxide, is built onto the silicon surface which is overlapped by photoresist. Positive photoresist becomes soluble when UV rays are exposed on it. Using proper alignment, the UV rays are passed through the masks which cast a shadow on the photoresist wherever the layer of silicon dioxide is required. Then the soluble photoresist is washed using appropriate solvent. Finally, the exposed silicon dioxide is etched away using chemicals and the remaining photoresist is removed to expose the regions of silicon dioxide that we required in our layer. The whole process is repeated for each layer.

Ion Implantation: Ions are accelerated at a very high energy and impinged on the target. The ion energy ranges from several KeV to MeV. The main purpose of ion implantation is doping in which impurities are added into wafer. This process is finalized with annealing process that repairs the lattice damage inflicted by high energy ions.

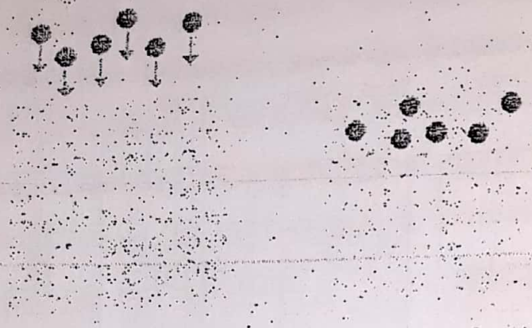

Figure 8.7: Ion Implantation and Lattice Damage after Implantation

Metallization: in this stage, a thin-film metal layer is produced which interconnects various circuit elements on the chip. Metallization also produces metalized area around the edge of the chip, also referred as bonding pads. Metal film can be deposited by physical vapor deposition (PVD) and chemical vapor deposition (CVD).

Wafer Testing: Number of ICs is produced in a single silicon wafer, which are subjected to test for errors or faulty ones. Testers or wafer probes are equipments used to test the correctness of the IC's by inspecting the output response for the streams of input.

- **Chip Cutout/Packaging:** Individual IC from wafer is cut out using a diamond scribe. Verified ones are mounted in an IC package which encapsulates the IC. Packaging prevents physical damage and corrosion, also supports electrical contact. Through hole package and surface mount package are examples of IC packaging. Single In-line Packaging and Dual In-line Packaging are types of through hole packaging.

PHOTOLITHOGRAPHY

Photolithography is the process which transfers a pattern from a mask to a light-sensitive chemical photoresist on the substrate. The word photolithography is from the Greek origin: photo means light, litho means stone and graphy means writing. It uses optical radiation to create patterns of complex circuit on a wafer. The various steps involved in photolithographic process are deposit barrier layer, photoresist coating, soft bake, mask alignment and exposure, develop photoresist, hard bake, etch window in barrier layer and remove photoresist.

The various steps of photolithography are explained below:

A. Deposit Barrier Layer

Barrier layers are the materials which are required to be laid on the substrate. It may be silicon dioxide, silicon nitride, poly-silicon, metals, etc. Different methods can be used for barrier formation: thermal oxidation, chemical vapor deposition, sputtering and vacuum evaporation. Silicon dioxide as a barrier layer is used to isolate one layer from another. For instance, it is used in electrical isolation of multilevel metallization. Silicon Dioxide can be grown using dry oxidation which uses O_2 gas in a chamber or wet oxidation in which the wafer is submerged in water. When heat is applied to the oxidation process, it increases the rate of SiO_2 growth.

B. Photoresist Coating

Photoresist is a substance which changes its characteristics when exposed to UV light. Before photoresist is coated, hexamethyldisilazane (HMDS) is used on the surface to improve adhesion. After that, liquid photoresist is coated over barrier layer using spin coating method. In this method, the wafer is held on vacuum chuck which is spun at about 3000-6000 rpm for about 15-30 seconds. Appropriate spinner rotational speed and viscosity of resist are essential factors to define photoresist's thickness which is about few micro-meters.

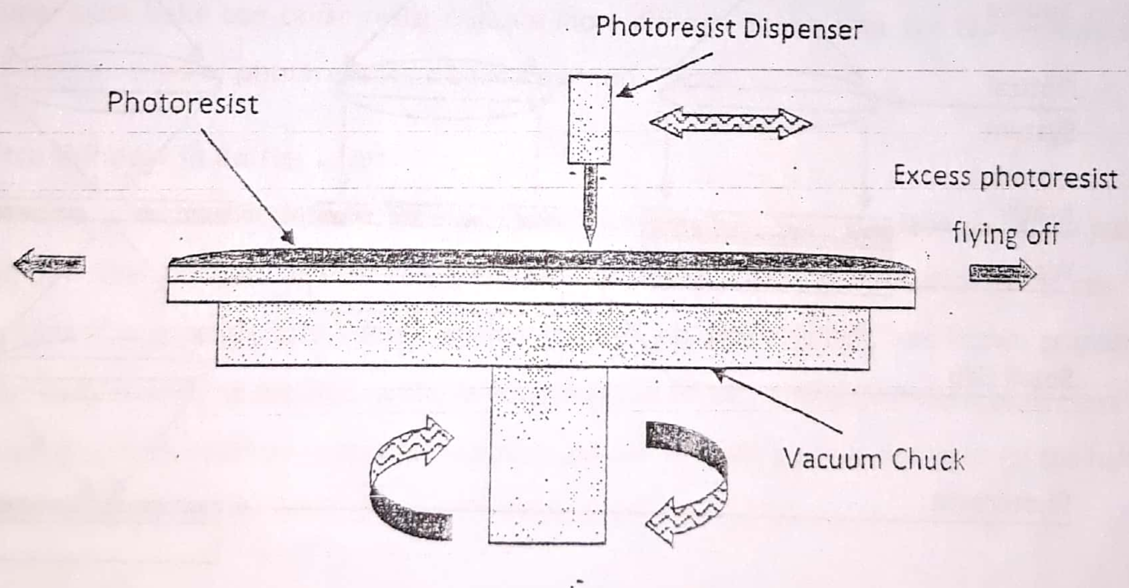

Figure 8.8: Photoresist Coating – Spin Coating Method

Types of Photoresist

Positive photoresist is insoluble in normal state but becomes soluble when exposed to UV light.

Negative photoresist is soluble in normal state but becomes insoluble when exposed to UV light.

C. Soft Bake or Pre Bake

Soft bake is simply the process of heating the wafer which removes the solvent from the photoresist. Baking time and temperature depend on the type of photoresist used and baking method. Different baking methods include hotplate, oven baking and microwave baking.

D. Mask Alignment and Exposure

Mask is simply an opaque plate with holes to pass UV rays. It contains pattern to be formed on wafer. Mask is aligned with the wafer accurately with the help of special device: steppers use automatic pattern recognition and alignment systems. Alignment masks are available on the mask and on wafer so as to make alignment more precise.

Once the mask has been precisely aligned, the photoresist is exposed through the pattern on the mask with a controlled amount of UV light. Exposure will cause exposed positive photoresist to become soluble whereas if negative photoresist is used then exposed part of it becomes insoluble. There are three primary exposure methods: contact, proximity, and projection.

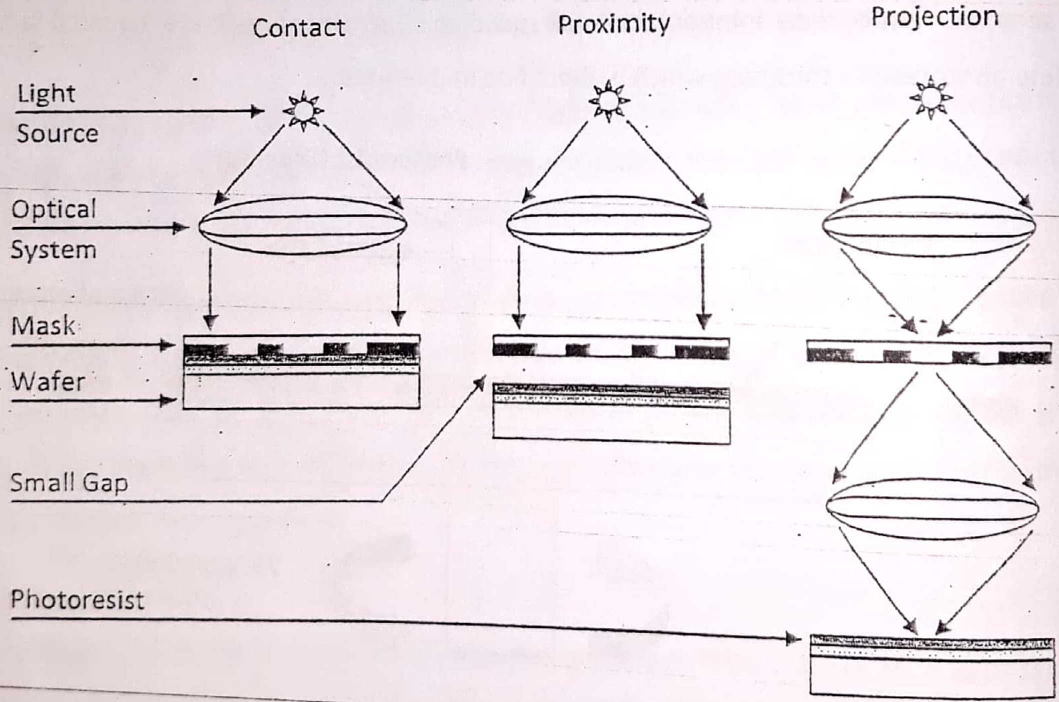

Figure 8.9: Different Exposure Methods

Contact Printing: In contact printing, the resist-coated silicon wafer and mask are brought into physical contact when exposed to UV light. This method results in very high resolution but the debris, trapped between the resist and the mask, can damage the mask and cause defects in the pattern.

Proximity Printing: In this method, small gap is maintained between wafer and the mask during exposure. The gap minimizes the risk of mask damage at the expense of resolution.

Projection Printing: in this printing method, an image of the patterns on the mask is projected onto the resist-coated wafer. High gap eliminates the risk of mask damage and high resolution is possible. For high resolution, only a small portion of the mask is imaged and stepped over the surface of the wafer.

E. Develop Photoresist

Barrier layer is exposed when the soluble photoresist is chemically washed away using a developer solution. In immersion develop method the photoresist-coated wafer is immersed in a developer solution. Then, it is rinsed with DI water and dried using spin dry method.

F. Hard Bake or Post Bake

Hard bake is used to stabilize and harden developed photoresist. It not only improves adhesion of the photoresist but also removes traces of solvent or developer solution. But, however, improper post bake can cause resist removal more difficult. Baking time and temperature can vary based on type of photoresist and baking method.

G. Etch Window in Barrier Layer

As hardened photoresist does not shield all part of barrier layer, etching method is implemented to remove the barrier layer which was left uncovered. Two methods of etching can be implemented: wet etch, also known as chemical etching, and dry etch, also known as plasma etching. In Wet etching method, wafer is submerged in HF acid and unprotected barrier layer is removed. Dry etch method uses plasma which collides with the surface and removes the layers of target material.

H. Remove Photoresist

Finally, the remaining photoresist is stripped from the surface exposing the required barrier layer. Photoresist can be removed by using solvent strippers, which cause the resist to swell and lose adhesion from the substrate. Another method of photoresist removal is by burning the resist in an oxygen plasma system and this process is called Resist Ashing.

The photolithography process can be summarized diagrammatically as:

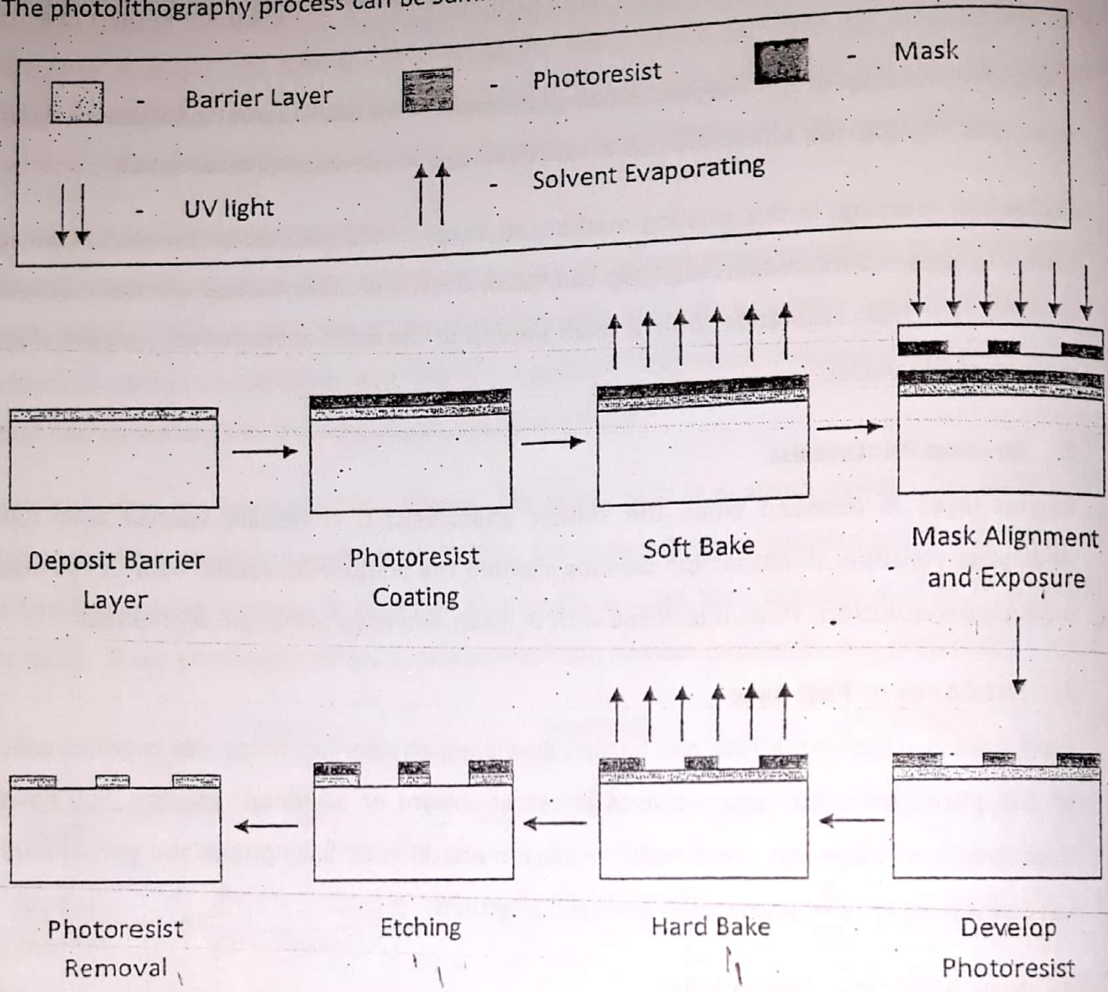

Figure 8.10: Various steps of photolithography process

8.2 Full-Custom (VLSI) IC Technology

Full-Custom IC technology includes VLSI (Very Large Scale Integrated Circuit) design in which the designer designs the complete transistor-level circuit for every processor, memory and other components used in the design. In this technology, first the designer creates layouts for basic components. And then, components are placed and connected, which are later translated to masks. Finally, the masks are given to the manufacturer for fabrication of IC of final design. The design steps are shown in the figure 8.11.

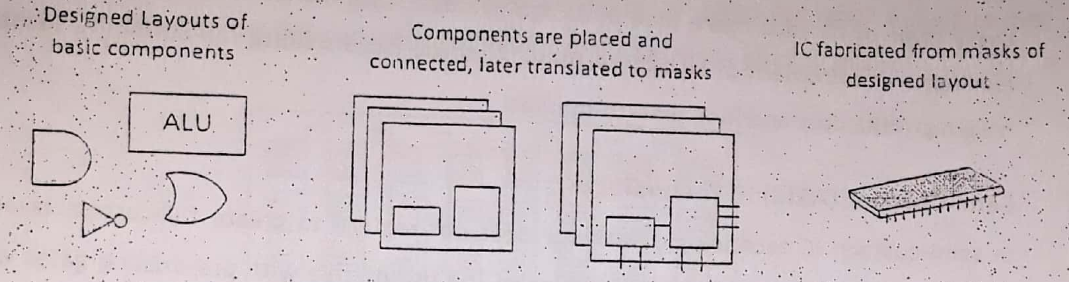

Figure 8.11: Full-Custom IC Technology

Placement, routing and sizing are few important physical design tasks that should be done carefully for an efficient layout design. Placement represents the task of placing and orienting the transistors on the IC. Routing is the task of connecting wires between the transistors. In sizing, width of each wire along with size of transistor is taken into considerations. Placement and routing should be done so as to avoid overlapping of transistors and wires. Placement also defines the length of wire required to connect transistors. Large size of wires and transistors provide better performances, but it increases power consumption and demands more silicon area in the IC. Compact layout can lead to an efficient design. For instance, transistor placed at closer distance requires shorter connecting wires, which further decreases the silicon size in the IC. In early days, compact designs were implemented using hand layout technique which is generally used for small and critical components. Today, however, physical design tools are used for automatic layout of the design which runs for hours or days to generate the optimized layout for better performance.

Advantages

Excellent efficiency: With respect to power consumption, performance and size, full-custom IC technology can be highly efficient. Since layout design is done by the designer, the components can be placed closer to each other which can be connected using short wires. Such layout yields optimum performance, size and power.

No wasted area and no unused transistors: In full-custom design, the required transistors for the circuit are placed on the IC. But there are no unused transistors which prevents wasted area.

Disadvantages

High NRE Cost and Long Time-to-market: Designing a complete layout, even with the help of CAD tools, can be time-consuming and prone to error. In addition to that, creating masks for every layer of IC adds more time in design process. Also, manufactured IC may contain errors leading to requirement of several re-spins. All these factors cause full-custom IC technology to have high NRE cost and long time-to-market.

8.3 Semi-Custom (ASIC) IC Technology

In semi-custom IC technology, designer does not require to create full-custom layout rather connects the pre-positioned building blocks. The use of chip with pre-existing gates will lessen the design work of layout and mask creation. So, the NRE cost is reduced while the time-to-market is relatively fast as compared to full custom IC technology. But, however, there will be a reduction in performance in terms of power, size and speed. Two types of semi-custom IC technologies are described in the following paragraphs.

Gate Array Semi-Custom IC Technology

In a gate array IC technology, a chip with arrays of pre-designed logic gates is utilized to implement the desired circuit. Here, the masks for transistor and gate levels are already designed, so the designer has the task of connecting pre-designed gates to achieve the desired implementation. In this technology, a set of masks of predefined gates are provided to the designer who then provides the connections among gates to implement require circuit. Masks of connections are generated and all masks are used to fabricate the IC.

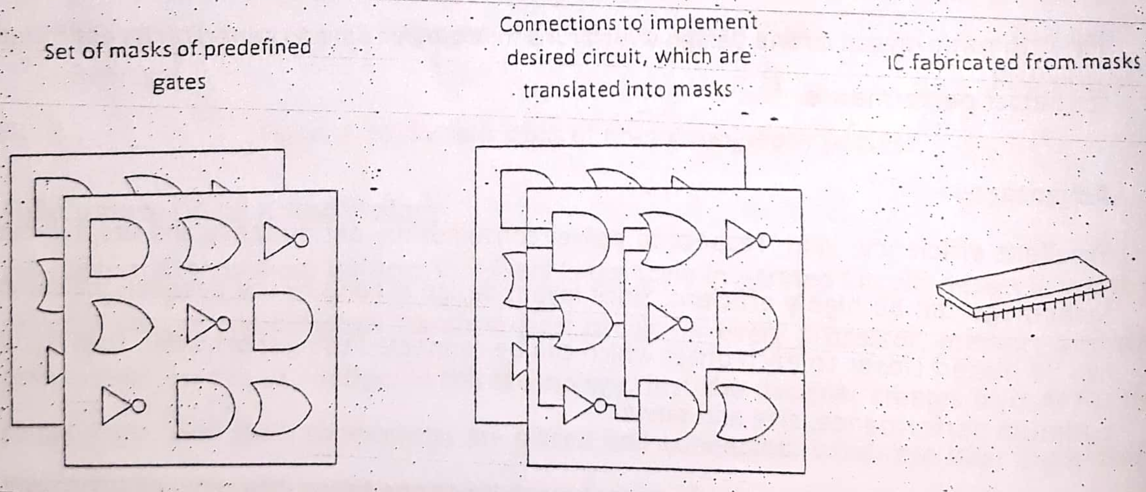

Figure 8.12: Gate Array Semi Custom IC Technology

This technology results in fast and relatively inexpensive design cycles. But, gates are placed in advance which may result in many unused gates, since all instances of each type of gate may not be required in our desired circuit. Also, the fixed placement of gates can result in long routing wires between gates as the connection is not known while gates are already placed.

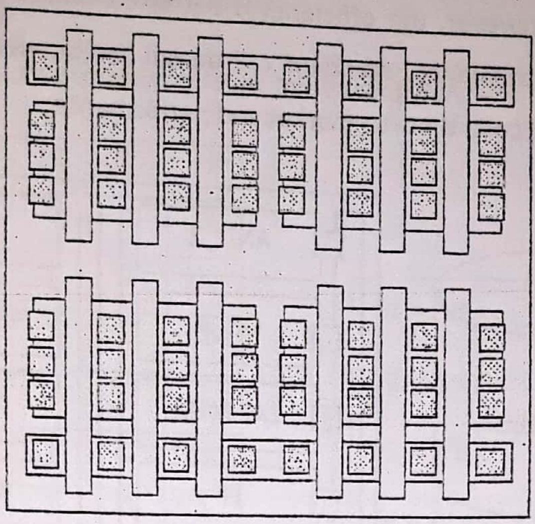

Figure 8.13: A simplified gate array layout

Standard Cell Semi-Custom IC Technology

In standard cell semi-custom IC technology, functional blocks, which are also called cells, with known electrical characteristics are utilized in the design to achieve very high gate density and good electrical performance. Cells may include logic gates such as NAND, NOR etc and other function blocks like multiplexor, flip-flop etc. In this technology, designers are facilitated with a library of predesigned cells from which the designer selects the required cells that are needed in the desired circuit. Masks of cells are created after the cells are placed and connected. Also the masks of connections among cells are generated. Using those masks, the IC is fabricated.

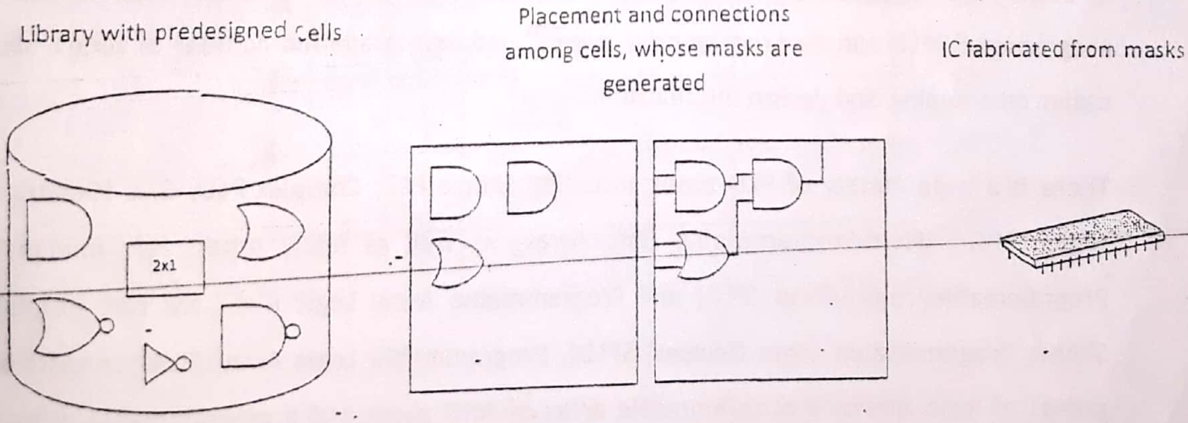

Figure 8.14: Standard Cell Semi-Custom IC Technology

The designer selects the cell, its position and its routing mechanism. So, it requires more NRE cost and longer time-to-market as compared to gate-array technology but still requires less than that of full-custom. However, the efficiency is better compared to gate array but less efficient than full-custom design. Hence, standard cell design lies between gate array and full custom design in terms of NRE cost, time-to-marker and performance.

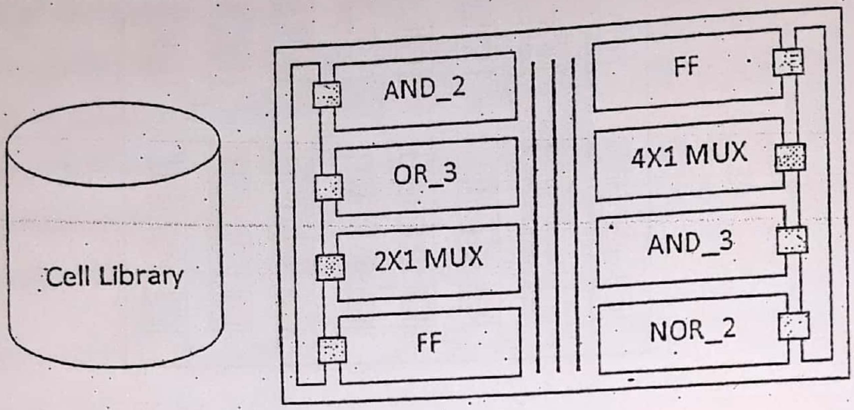

Figure 8.15: A simple standard cell layout

8.4 Programmable Logic Device (PLD) IC Technology

In programmable logic device IC technology, there exist programmable circuits which are programmed by the designer to implement the required design. Programming, in this case, means creating or breaking connections between wires that connect gates, either by blowing a fuse with high current, or setting a bit in a programmable switch. In this technology, a pre-fabricated chip with no logic function programmed is made available to the designer who then programs the required portions of the chip to implement the desired functionality.

It offers the designer the facility of changing design functions even after it has been programmed. PLD can be programmed, erased, and reprogrammed number of times, allowing easier prototyping and design modification.

There is a wide variety of PLD types, including Simple PLD, Complex PLD, GAL (Generic Array Logic), FPGA (Field-Programmable Gate Array) as well as many others left unmentioned. Programmable Logic Array (PLA) and Programmable Array Logic (PAL) are two examples of Simple Programmable Logic Devices (SPLD). Programmable Logic Array (PLA) consists of two planes of logic arrays: a programmable array of AND gates and a programmable array of OR gates. The AND plane and the OR plane give the possibility to computer any function expressed as a sum of products. Every AND gate in AND plane is associated with inputs and complement of inputs to generate any product term. And, OR gate generates the sum of AND gate outputs. The example of PLA is shown in the figure below.

Example 2: Implement the following truth table using PLA

A	B	C	F1	F2	F3	F4
0	0	0	0	0	1	1
0	0	1	0	1	0	1
0	1	0	0	1	0	1
0	1	1	0	1	0	1
1	0	0	0	1	0	1
1	0	1	0	1	0	1
1	1	0	0	1	0	1
1	1	1	1	1	0	0

F1 = ABC

F2 = A + B + C

F3 = A'B'C'

F4 = A' + B' + C'

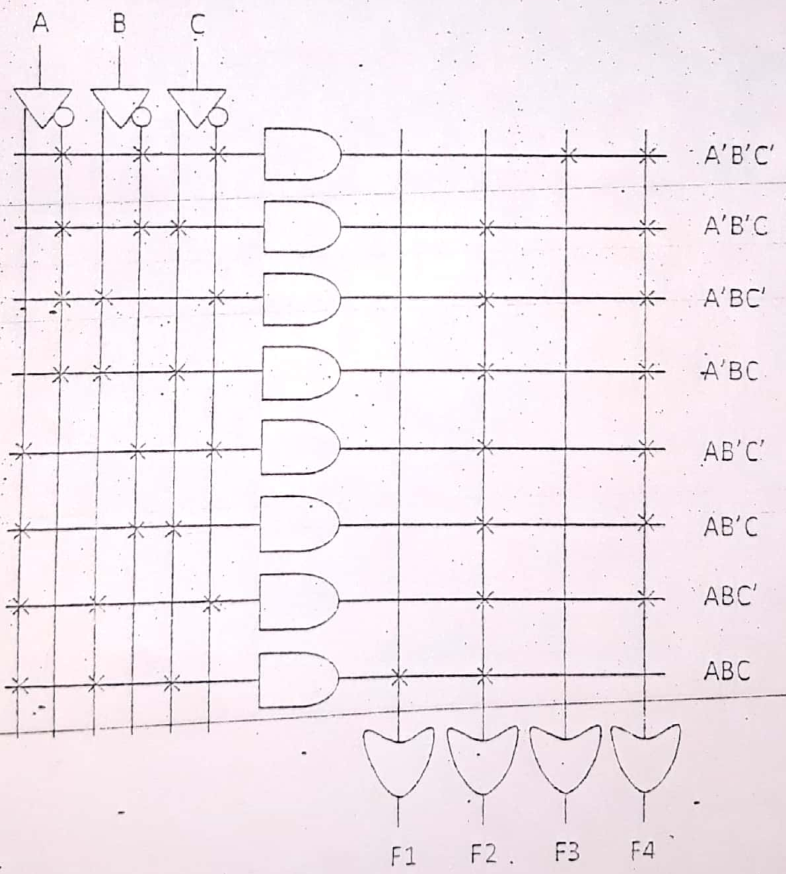

Figure 8.16: PLA implementation for given truth table

Programmable Array Logic uses just one programmable array: fixed OR matrix and programmable AND matrix. It decreases number of expensive programmable components which further reduces size and delay. PLA and PAL are generally used for low-complexity problems which require fairly high speed. As the complexity grows, Complex Programmable Logic Devices (CPLD) must be used. CPLDs are the integration of numerous SPLDs with added programmable interconnect between them. CPLD is a combination of fully programmable AND/OR array which perform a multitude of logic functions and microcells which perform combination or sequential logic. CPLDs may use analog sense amplifiers to boost the performance but at the cost of very high current requirements.